COURSE KICK-OFF

Building end-to-end agentic applications on Databricks

12 October - 29 November 2026

7 weeks of hands-on LLMOps



Your teachers



Maria Vechtomova

Core instructor

Co-founder of Cauchy
Databricks MVP
12+ years in Data & AI



Basak Eskili

Core instructor

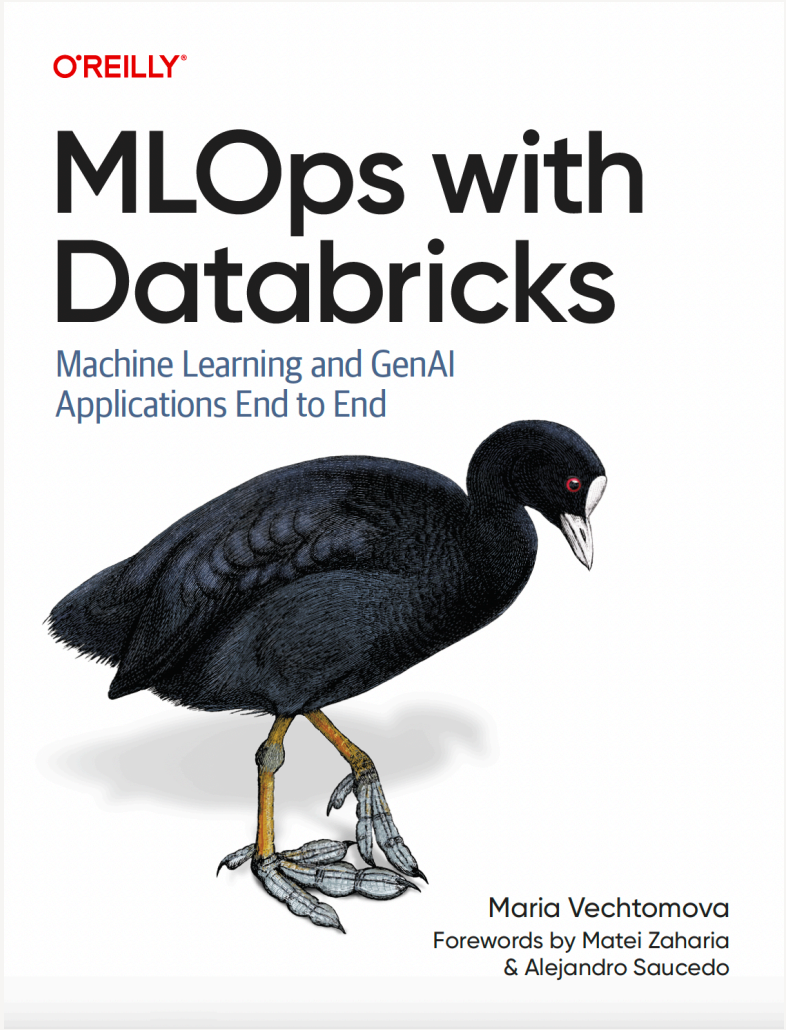
ML Engineer at Booking.com
MLOps and production ML
10+ years in Data & AI



Victor Kuznetsov

Course assistant

Databricks Champion
Data Engineer
10+ years in Data & AI



MLOps with Databricks

Machine Learning and GenAI Applications End to End

Maria Vechtomova

Learning goals

An end-to-end application you can explain, evaluate, and operate

LLMOps foundations

Understand the operational challenges of LLM workflows and agents.

Agents and retrieval

Build a domain-specific application using your data and tools.

Evaluation and deployment

Trace behavior, evaluate quality, and deploy your application.

Production practices

Apply CI/CD, monitoring, and governance to your own project.

How we learn

A shared example	We teach the concepts through an arXiv paper curator agent on Databricks.
Your own use case	Apply each week's concepts to an application you want to build.
Weekly PR reviews	Submit your work through GitHub pull requests for structured feedback.
Operations throughout	Agent design matters. Our focus is the system around the agent.

The weekly rhythm

WEDNESDAY SESSIONS

16:00-18:00

Amsterdam time

Recordings after each session

MONDAY MEETINGS

12 October

Course kick-off

23 November

Platform infrastructure introduction

Q&A throughout the week

Use Discord for Q&A throughout the week.
If we need extra Q&A, we can schedule a session.

Amsterdam: UTC+2 through 24 October, UTC+1 from 25 October. Calendar invites show your local time.

Curriculum

12 October - 29 November 2026

01
12-18 October
LLMOps Foundations & developing on Databricks
Mon 12: kick-off · Wed 14: teaching session

02
19-25 October
Data & knowledge processing
Wed 21 October

03
26 October - 1 November
MCP. Tool calling. Agent orchestration
Wed 28 October

04
2-8 November
MLflow for GenAI
Wed 4 November · Reflection due Thu 5

05
9-15 November
Deployment strategies for agents & DABs
Wed 11 November

06
16-22 November
Post-deployment: running AI in production
Wed 18 November

07
23-29 November
Platform infrastructure introduction
Mon 23 Nov

Demo day: show your project!
Wed 25 Nov

Your course access

Your starting point, unless we provided separate instructions to you



GitHub repository

Your dedicated repo

Accept the invitations for both the repository and the organization. Submit your project work as pull requests.

Repo + organization invite



Databricks access

Your course environment

You have access to our Databricks workspace and your own dashboard. We will walk through these together.

Workspace + personal dashboard



Discord

Private course channel

Ask questions, share progress, and get help throughout the week. We can schedule extra Q&A if needed.

Questions throughout the week

Course calendar, materials, and recordings: Maven

Your environment, live

Screen-sharing walkthrough · unless we provided separate instructions to you

01 Open Databricks and find your workspace

02 Open your personal dashboard and review what is available

03 Find your GitHub repository and the pull request workflow

04 Locate the calendar, materials, recordings, and Discord channel

Follow along with your own account. Flag any missing access during the walkthrough.

Materials and support

Reading on Maven

Slides, lecture transcriptions, and reading material.

Recordings on Maven

Session recordings available after each lecture.

Your progress

Use the course dashboard and Genie to track your performance.

Code feedback

Weekly GitHub PR reviews

Q&A

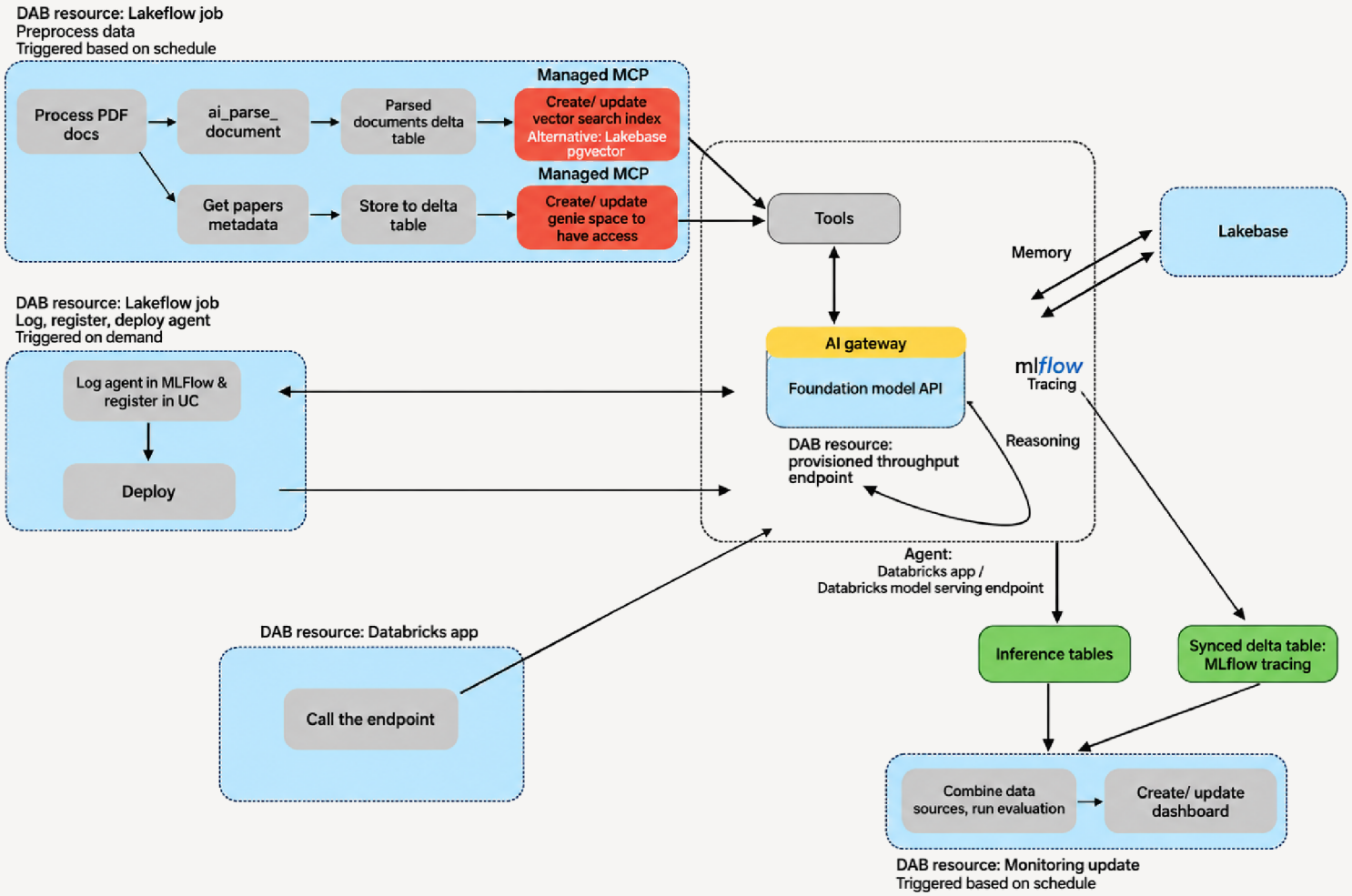
Discord throughout the week

Q&As can be scheduled in case of demand.



The project we demonstrate

ArXiv curator



Your project

A use case you will develop throughout the course

A clear problem

Identify who will use the application and what it should help them do.

Data and tools

Choose the sources and capabilities your application needs.

Evidence of quality

Show traces and evaluation results alongside the application.

Demo day · 25 November

Demonstrate what you built and explain the operational decisions.



Before Wednesday

- Verify your Databricks access
- Accept your GitHub invitations
- Join Discord and check your calendar
- Bring an initial idea for your course project
- Fill in the form

  Unless we provided different instructions to you

Wednesday 14 October, 16:00 Amsterdam



MLOps vs LLMOps

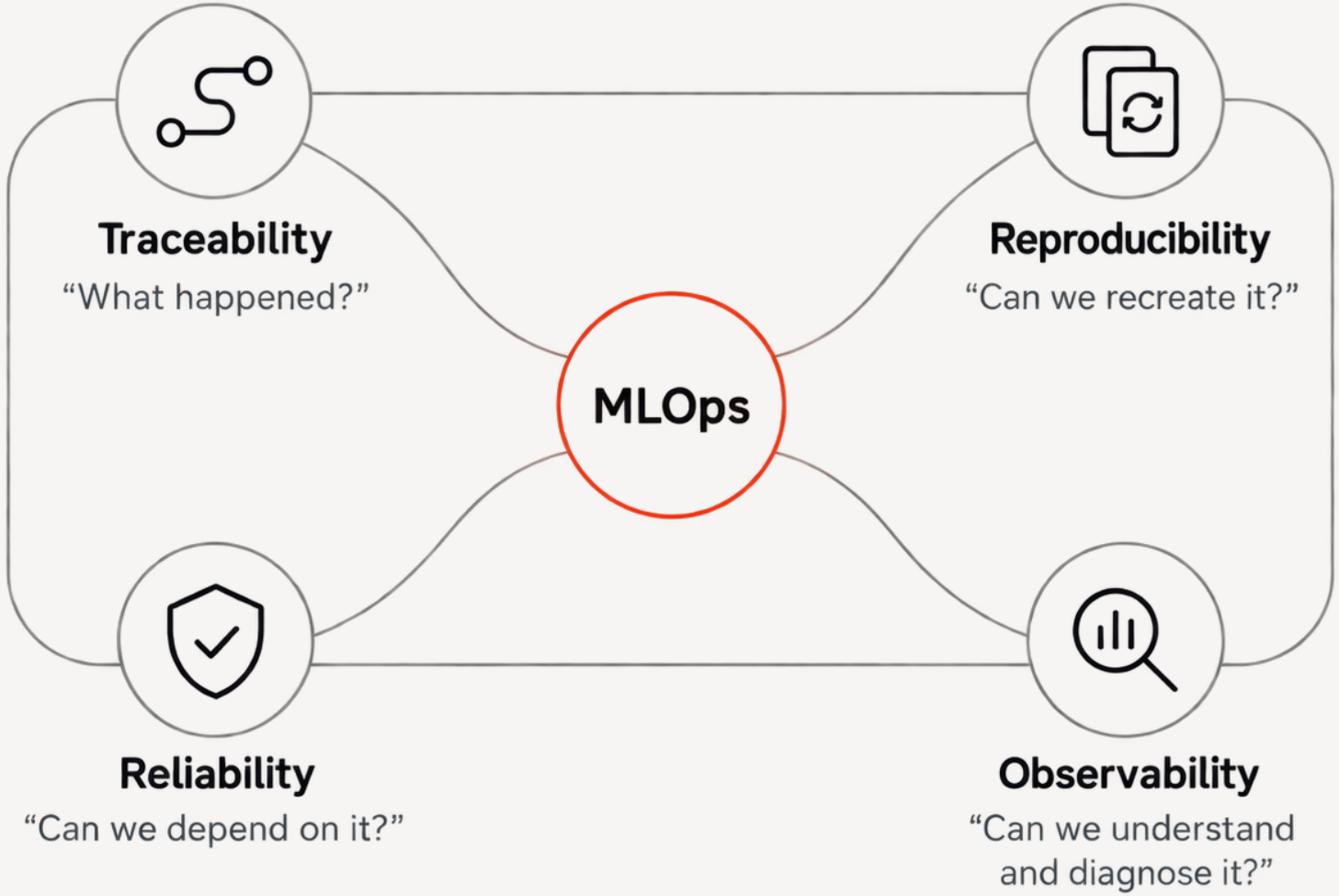


> MLOps is the set of practices for deploying and operating machine learning models reliably and efficiently.
>
> LLMOps extends these practices to LLM workflows and agents.

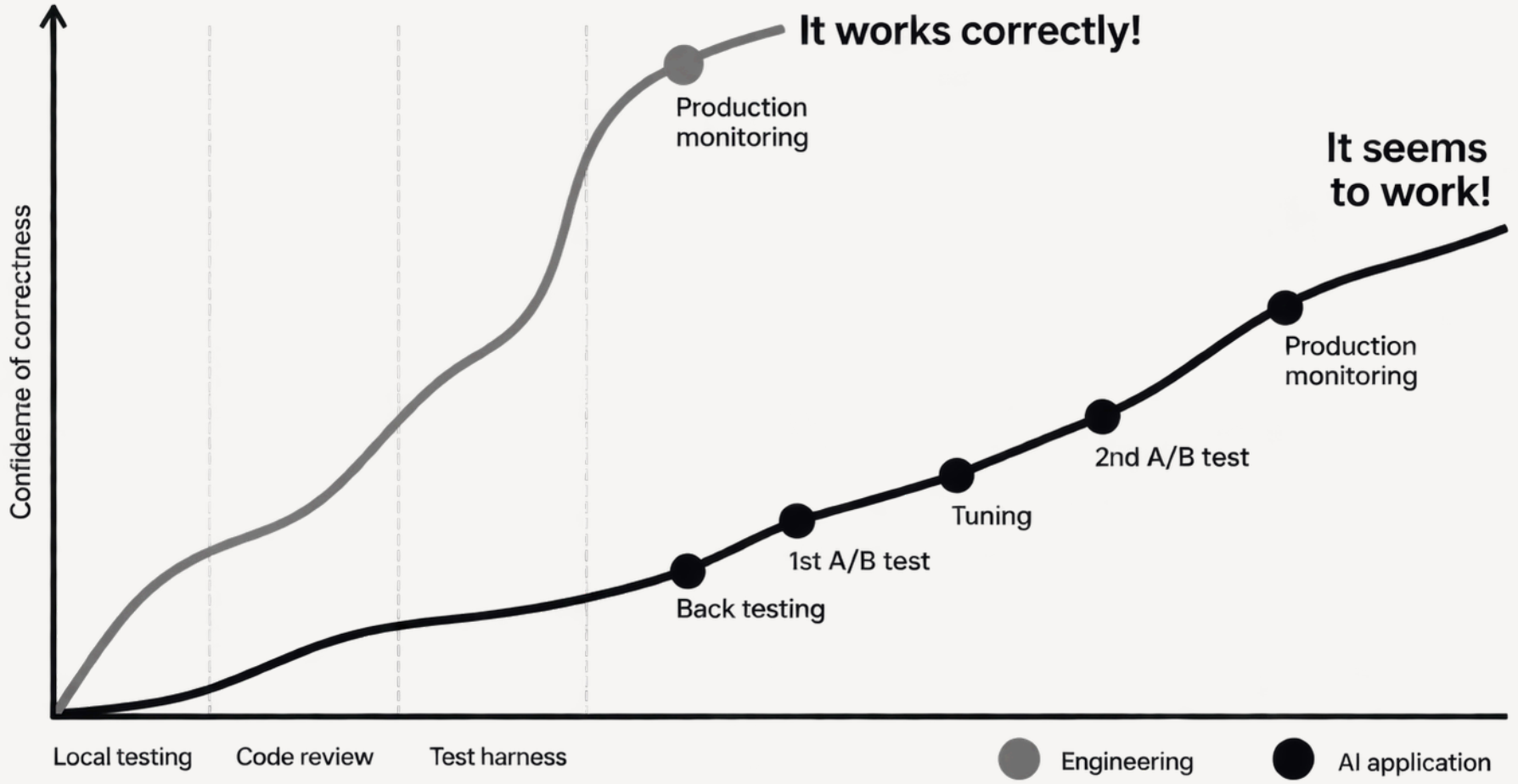


MLOps principles

These principles reinforce one another.



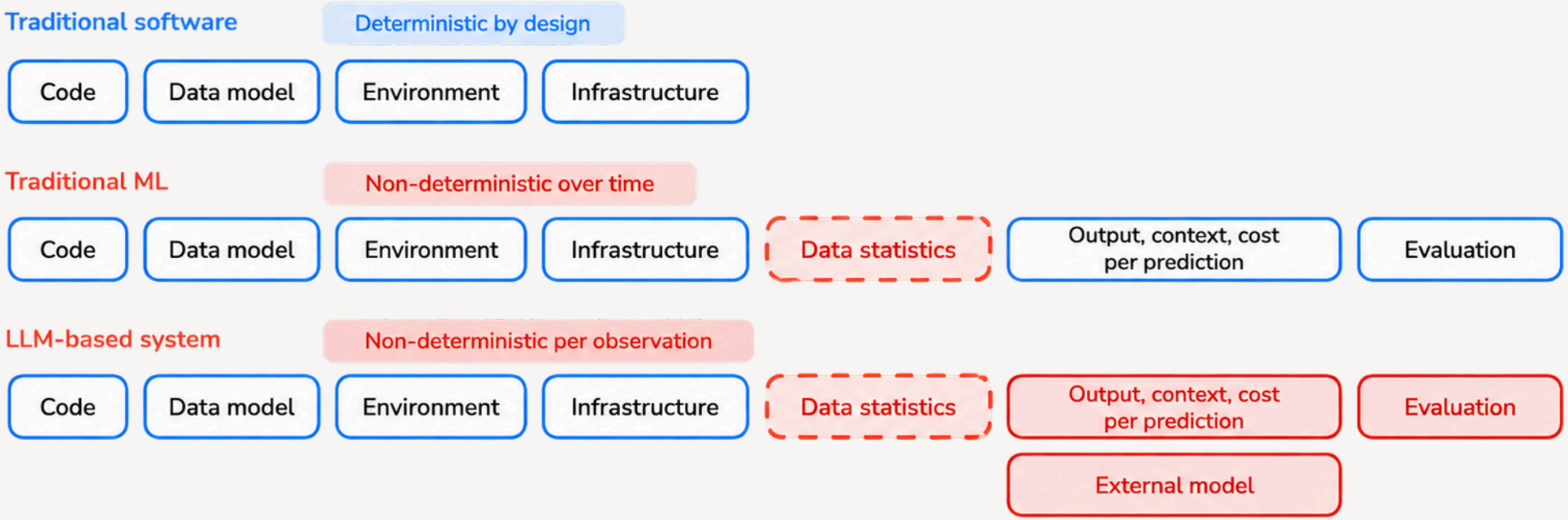
System lifecycle starts after deployment



The Complexity Ladder

Traditional software > Traditional ML > LLM-based system

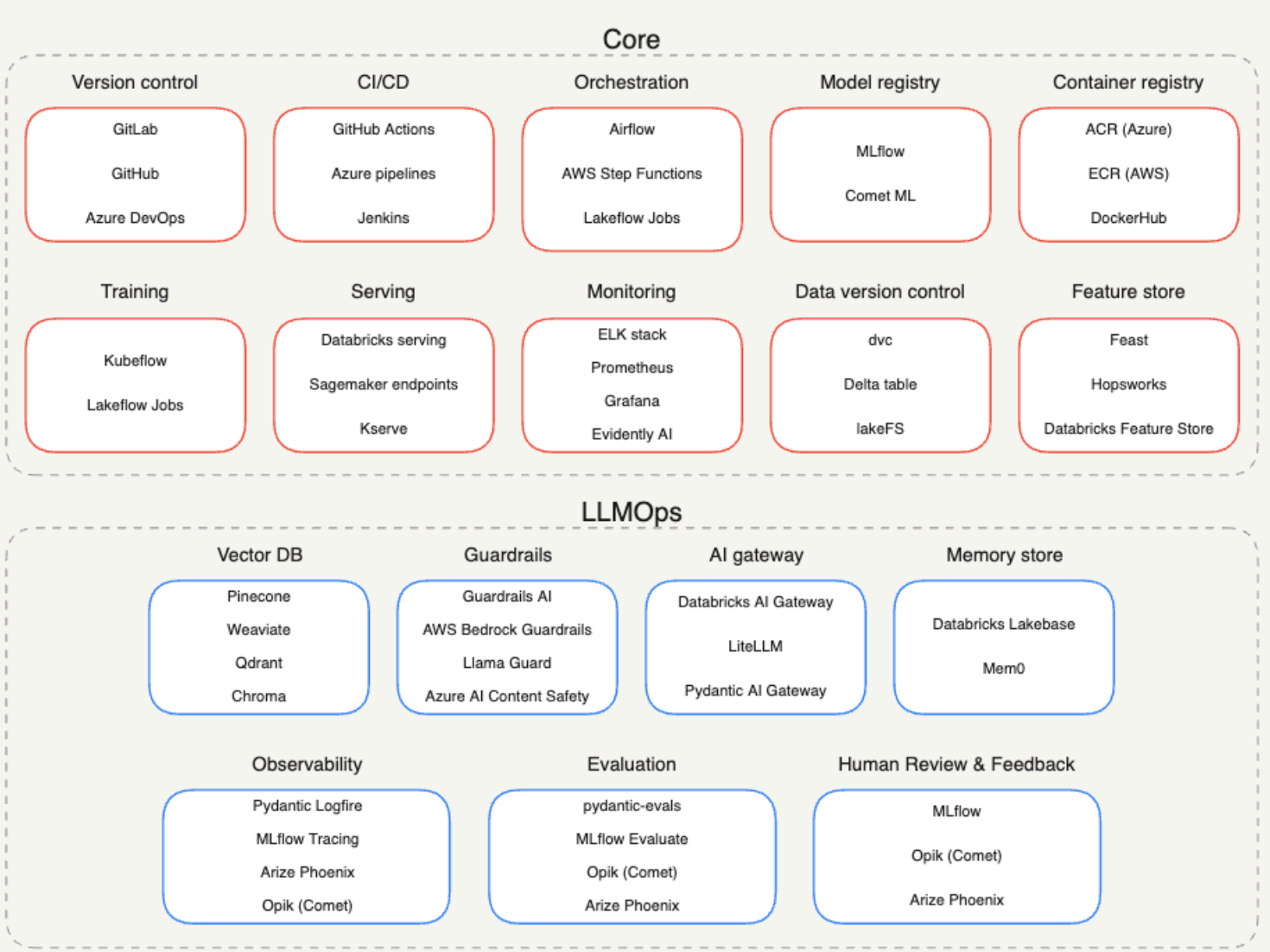
Each step inherits the complexity of the previous one and adds a new source of non-determinism.



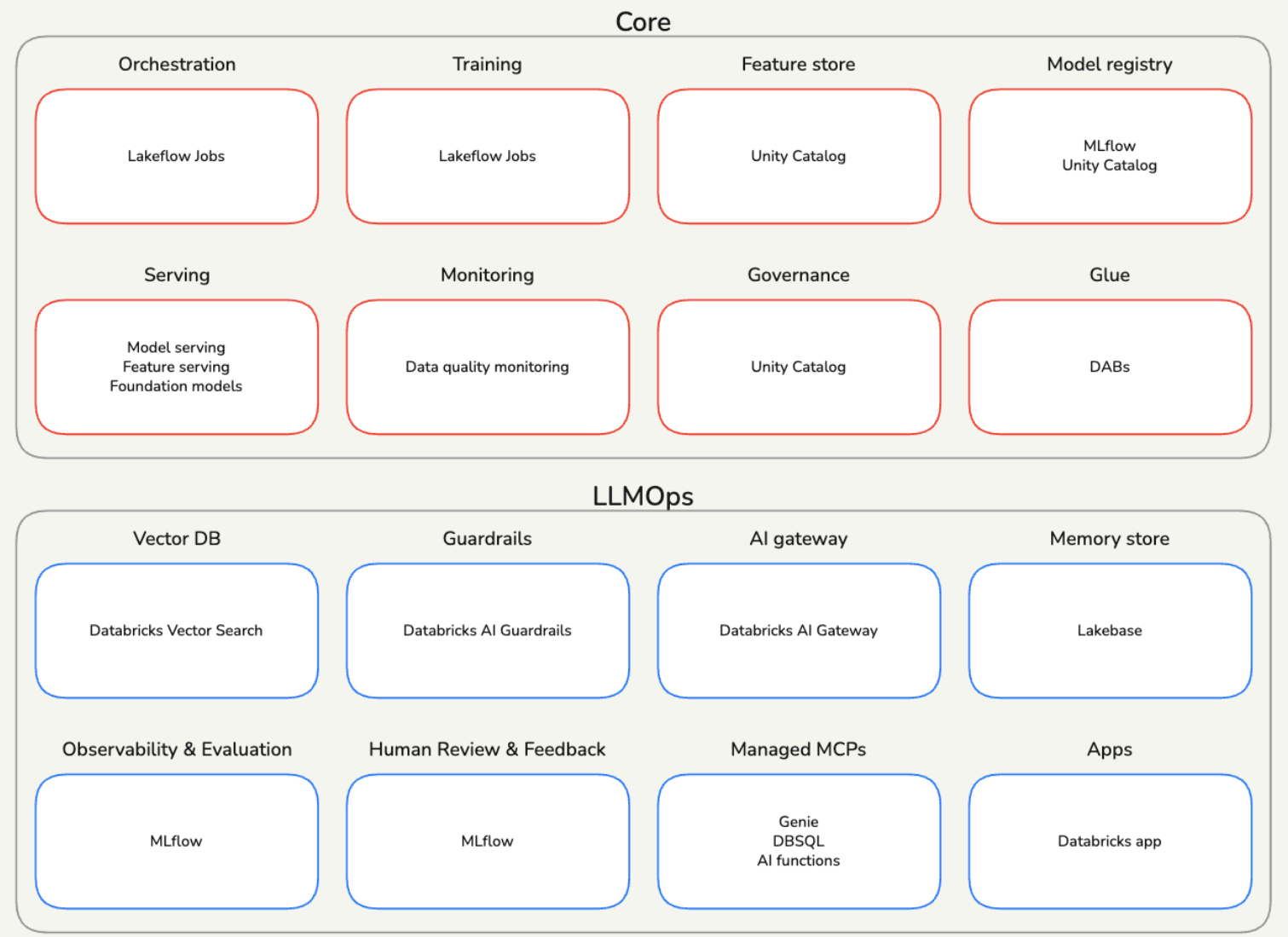
Key insight: A software system with identical properties produces identical output. An ML system cannot guarantee this. An LLM system adds further uncontrollable variables: an external model you do not own, plus non-deterministic output, retrieved context, cost, and even evaluation.

The rungs are not clean steps: deep learning and NLP already blur them with GPU non-determinism and evaluation that was never fully objective. LLMs remove the last guarantees.

The wider tooling landscape



The Databricks ecosystem



Questions?

Course format, access, or your project idea

After today, continue the conversation on Discord.



Thank you

See you on Wednesday!

14 October · 16:00 Amsterdam

